~~Advanced~~ Number Theory 2

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Goals

- Modular Arithmetic (Revision)
- Binary Exponentiation (Revision)
- Euclidean Algorithm
- GCD Properties
- Euler’s Theorem
- Fermat’s Theorem
- Mod Inverse under Euler’s & Fermat’s Theorem

Modular Arithmetic [Rev.]

10^{1000}

ans % MOD

- $(A + B) \% M = [(A \% M) + (B \% M)] \% M$???
- $(A - B) \% M = [(A \% M) - (B \% M) + M] \% M$
- $(A * B) \% M = [(A \% M) * (B \% M)] \% M$
- $(A / B) \% M = [(A \% M) * (B^{-1} \% M)] \% M$

What is B^{-1} % M ? Modular Inverse (Will be covered later)

$\frac{a}{b} \rightarrow a * \frac{1}{b} \rightarrow a * b^{-1}$

Imp**

$$\underbrace{a \% b}_{\text{repeated subtraction}} \rightarrow [0 \ldots b-1]$$

$$15 \rightarrow 13 \rightarrow 11 \rightarrow 9 \rightarrow 7 \rightarrow 5 - 3 - 1$$

$$(a+b)\ \%\ m$$

$$xm + r_a$$

$$ym + r_b$$

$$\Big(\underbrace{\cancel{xm}}_{0} + \underbrace{r_a}_{a\%m} + \underbrace{\cancel{ym}}_{0} + \underbrace{r_b}_{b\%m}\Big)\ \%\ m$$

Binary Modular Exponentiation [Rev.]

Idea:

$$a^n = \begin{cases} 1 & \text{if } n == 0 \\ \left(a^{\frac{n}{2}}\right)^2 & \text{if } n > 0 \text{ and } n \text{ even} \\ \left(a^{\frac{n-1}{2}}\right)^2 \cdot \underline{a} & \text{if } n > 0 \text{ and } n \text{ odd} \end{cases}$$

Time Complexity: O(log(n))

Binary Exponentiation

$a^b \longrightarrow$

```
prod = 1
for(i=1; i<=b ; i++) {
    product *= a
}
```

Time complexity : $O(b)$

$\underline{\log_2 b}$ values

b

$$a^{7} = a^{\overset{2^2\,2^1\,2^0}{111}\,(2)} = a^{(2^0+2^1+2^2)} = a^1 \cdot a^2 \cdot a^4$$

a^4

$$a^{5} = a^{101} = a^{(2^0+2^2)} = a^1 \cdot a^4$$

$$\log_2 b$$

$$a^1 \quad a^2 \quad a^4 \quad a^8$$

$$\boxed{O(\log_2 b)}$$

$$a^1 \xrightarrow{(a^1)^2} a^2 \xrightarrow{(a^2)^2} \underline{a^4} \xrightarrow[a^4 \cdot a^4]{} a^8$$

$res = \underline{1 \times a^{16}} \rightarrow a^{16}$

$a^{\cancel{32}}$

Binary Modular Exponentiation [Rev.]

Implementation: (Don't forget to take MOD when mentioned)

Recursive

```
long long binpow(long long a, long long b) {
    if (b == 0)
        return 1;
    long long res = binpow(a, b / 2);
    if (b % 2)
        return res * res * a;
    else
        return res * res;
}
```

Iterative

```
long long binpow(long long a, long long b) {
    long long res = 1;
    while (b > 0) {
        if (b & 1)
            res = res * a;
        a = a * a;
        b >>= 1;
    }
    return res;
}
```

9^{10^9} → For loop → TLE

→ 30 iterations!

Euclidean Algorithm

$gcd(x, 0) = x$ magical formula?!

Theorem: $$\gcd(a,b) = \begin{cases} a, & \text{if } b = 0 \\ \gcd(b, a \bmod b), & \text{otherwise.} \end{cases}$$

Implementation:

Recursive

```
int gcd (int a, int b) {
    if (b == 0)
        return a;
    else
        return gcd (b, a % b);
}
```

Iterative

```
int gcd (int a, int b) {
    while (b) {
        a %= b;
        swap(a, b);
    }
    return a;
}
```

Time Complexity: O(log(min(a, b)))

$$\underbrace{\gcd(a,b)}_{\hookrightarrow g} \rightarrow \text{greatest common divisor}$$

$$\gcd(a,b) \rightarrow g$$

$$a = \overset{2}{x}\,g \cdot 2$$
$$b = \underset{2}{y}\,g \cdot 2$$

* x & y are coprime

$a = xg$

$b = yg$

$a > b$

$(a-b) \to$ divisible by g

$$\gcd(a, b) = \gcd(\underline{b}, \underline{a-b})$$

$\gcd(2, 17)$

$\hookrightarrow \gcd(2, 17-2) \to \gcd(2, 15)$

$\hookrightarrow \gcd(2, 15-2)$

$\hookrightarrow \gcd(2, \underline{13})$

smaller than 2

$\hookrightarrow \gcd(2, 1)$

gcd (17, 2)

↳ gcd (1, 2) → gcd (2, 1)

↳ gcd (0, 1)

** Hard

$$\boxed{\underline{a \% b} \le \frac{a}{2} \qquad a \ge b}$$

[it gets atleast halfed]

Proof: ?!

1. $\underbrace{a > 2b}$

↳ a loses half its value when $a \% b$

2. $\underbrace{b < a \leq 2b}$ $\qquad a \% b$

$\underline{\cancel{b}+1} \qquad \underline{\cancel{b}+2} \qquad \ldots\ldots \qquad \underline{\cancel{b}+\cancel{b}}$

0

→ Atleast getting halved

Important GCD Results

- GCD(a, b) = GCD(b, a)
- GCD(a, 0) = a
- GCD(a, b, c) = GCD(GCD(a, b), c) = GCD(a, GCD(b, c)) = GCD(b, GCD(a, c))

  gx gy gz

- GCD(a, b) >= GCD(a, b, c) >= GCD(a, b, c, d)
- GCD contains the minimum powers of primes
- LCM contains the maximum powers of primes
- GCD(a, b) * LCM(a, b) = a * b

$$* \ LCM(a,b) = \left[\frac{a * b}{gcd(a,b)}\right]$$

$$18 = 2^1 \cdot 3^2$$
$$36 = 2^2 \cdot 3^2$$

$$\underbrace{2^1}\times\underbrace{3^2} \xrightarrow{GCD} 18$$

$$2^2 \cdot 3^2 \xrightarrow{LCM} 36$$

Euler's Totient Function

phi(N) = number of values X such that X <= N and GCD(X, N) = 1
phi(N) is a multiplicative function.

The following properties of Euler totient function are sufficient to calculate it for any number:

- If p is a prime number, then $\gcd(p, q) = 1$ for all $1 \leq q < p$. Therefore we have:

$$\phi(p) = p - 1.$$

- If p is a prime number and $k \geq 1$, then there are exactly p^k/p numbers between 1 and p^k that are divisible by p. Which gives us:

$$\phi(p^k) = p^k - p^{k-1}.$$

- If a and b are relatively prime, then:

$$\phi(ab) = \phi(a) \cdot \phi(b).$$

$\varphi(n) \rightarrow$ number of coprime numbers between $1 - n$.

1. n is prime (p)

$\hookrightarrow \varphi(p) = p-1$

2. n is $(\text{prime})^k$ $[1 - p^k]$

$\hookrightarrow \varphi(p^k)$

$[1 - 5^{10}]$

not divisible by 5 should be considered

$[1 - 5^{10}]$ $\qquad \dfrac{5^{10}}{5}$ gcd $\neq 1$

$5^{10} - \dfrac{5^{10}}{5}$ $\qquad p^k - p^{k-1}$

$$\varphi(n) = \varphi(p_1^{k_1}) \cdot \varphi(p_2^{k_2}) \cdots$$

$$= (p_1^{k_1} - p_1^{k_1 - 1}) \cdot (p_2^{k_2} - p_2^{k_2 - 1}) \cdots$$

$$= p_1^{k_1}\left(1 - \frac{1}{p_1}\right) p_2^{k_2}\left(1 - \frac{1}{p_2}\right) \cdots$$

$$\varphi(n) = n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdots$$

$$n \le 10^6 \longrightarrow \varphi(n)$$

1. $\underbrace{O(\sqrt{n})}$ $\rightarrow$ $\underbrace{n \le 10^{12}}$

2. $O(\underbrace{\log_2 n}_{\text{per query}} + \underbrace{\text{SPF precomp}}_{O(n \log \log n)})$

Euler's Totient Function

Idea:

$$\phi(n) = \phi({p_1}^{a_1}) \cdot \phi({p_2}^{a_2}) \cdots \phi({p_k}^{a_k})$$

$$= \left({p_1}^{a_1} - {p_1}^{a_1-1}\right) \cdot \left({p_2}^{a_2} - {p_2}^{a_2-1}\right) \cdots \left({p_k}^{a_k} - {p_k}^{a_k-1}\right)$$

$$= p_1^{a_1} \cdot \left(1 - \frac{1}{p_1}\right) \cdot p_2^{a_2} \cdot \left(1 - \frac{1}{p_2}\right) \cdots p_k^{a_k} \cdot \left(1 - \frac{1}{p_k}\right)$$

$$= n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right)$$

Euler’s Totient Function

Idea:

$$\phi(n) = \phi({p_1}^{a_1}) \cdot \phi({p_2}^{a_2}) \cdots \phi({p_k}^{a_k})$$

$$= \left({p_1}^{a_1} - {p_1}^{a_1-1}\right) \cdot \left({p_2}^{a_2} - {p_2}^{a_2-1}\right) \cdots \left({p_k}^{a_k} - {p_k}^{a_k-1}\right)$$

$$= p_1^{a_1} \cdot \left(1 - \frac{1}{p_1}\right) \cdot p_2^{a_2} \cdot \left(1 - \frac{1}{p_2}\right) \cdots p_k^{a_k} \cdot \left(1 - \frac{1}{p_k}\right)$$

$$= n \cdot \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right)$$

$$\left[a^{\varphi(m)} \% \underline{m} \equiv 1 \% m\right]$$

prime

when $m \to$ prime fermat's little theorem

$$\hookrightarrow \left[a^{m-1} \% m \equiv 1 \% m\right]$$

Made with Goodnotes

$$a^{m-1} \% m \equiv 1 \% m$$

$$a^{m-2} \% m \equiv \frac{1}{a} \% m$$

$$\boxed{a^{m-2}} \,\%\, m \equiv \boxed{\frac{1}{a}} \,\%\, m$$

Bin exponentiation

$$\left\langle \frac{a}{b} \,\%\, m = \left(a \,\%\, m * b^{m-2} \,\%\, m\right) \% \, m \right\rangle$$

where m is prime

Can we say this?

WRONG

$$\cancel{a^{b} \% m \equiv a^{b \% m} \% m}$$

where m is prime

✓ Right

$$\boxed{a^{b} \% m \equiv a^{\frac{b \% (m-1)}{}} \% m}$$

$\varphi(m)$

$$a^{b} \% m \equiv a^{b \% \varphi(m)} \% m$$

$$a^0 \% m \equiv 1 \% m$$

$$a^1 \% m$$

$$a^{m-1} \% m \equiv 1 \% m$$

$$a^m \% m$$

0 $\qquad\qquad\qquad$ $m-2 \rightarrow m-1$

$m-1$ $\qquad\qquad\qquad$ $2^{(m-1)}-1$

$a^0 \longrightarrow 1$

$a^1 \longrightarrow a^0 \cdot a$

a^2

$a^{m-1} \longrightarrow 1$

$a^m \% m \longrightarrow a^{m-1} \cdot a \to a^0 \cdot a \% m$

a^{m+1}

Euler's Theorem and Fermat's Theorem:

Euler's Theorem $a^{\phi(m)} \equiv 1 \pmod{m}$ if a and m are relatively prime.

Fermat's Theorem $a^{m-1} \equiv 1 \pmod{m}$

Practice Problem:

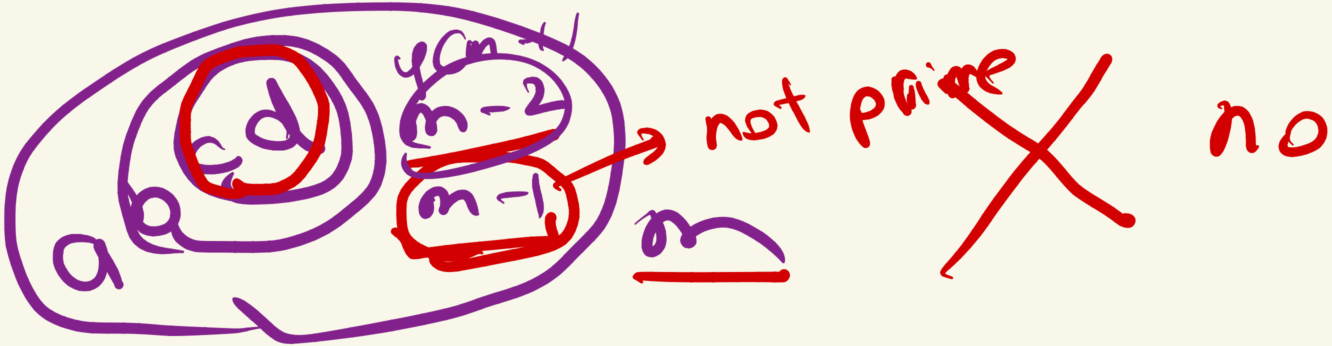
https://cses.fi/problemset/task/1712

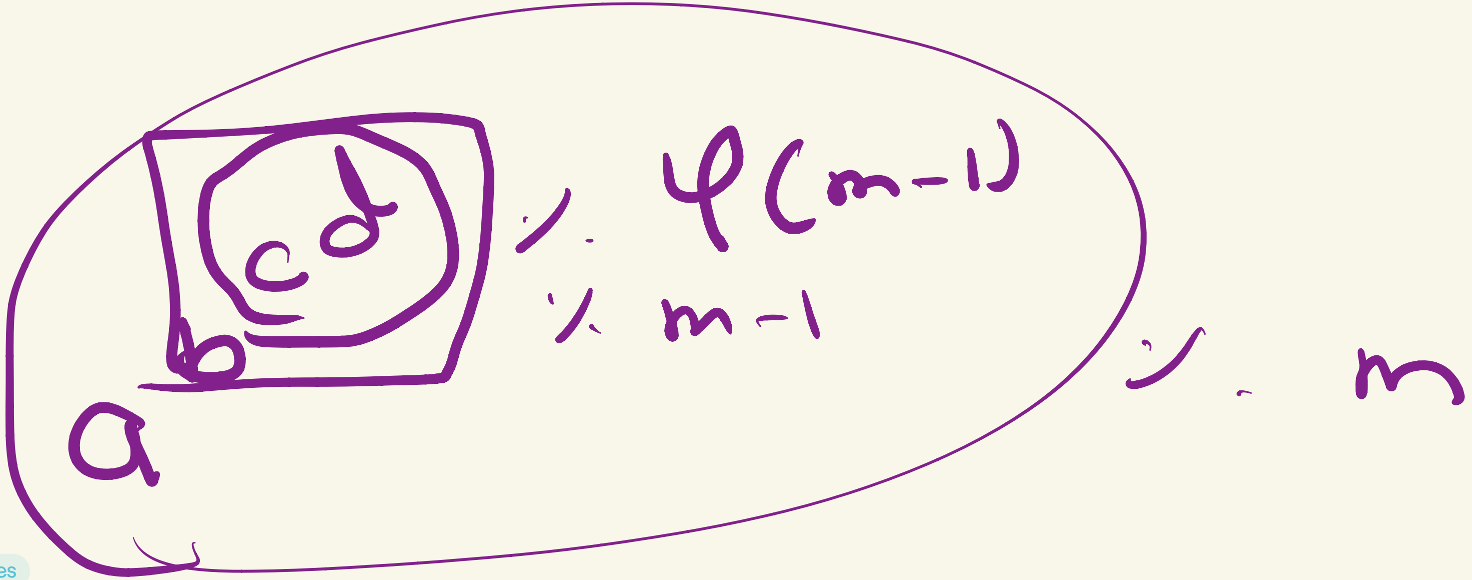
$a^{b^{c}} \rightarrow \varphi(p) \rightarrow p-1$

$\% \; p$

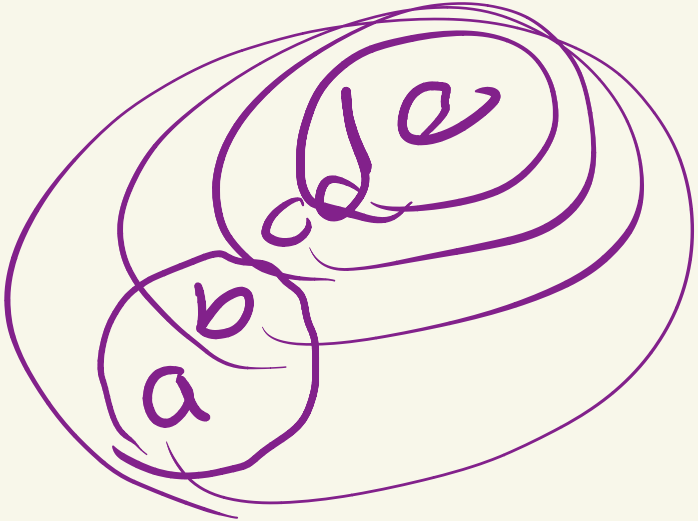
$$(a)^{\boxed{b^{c}}^{x}} \rightarrow \% \; p-1$$

$\hookrightarrow \underbrace{a^{x}} \; \% \; p$


a
b
c
d
φ(m-1)
m-2
m-1
m
not prime
no


a
b
c
d
% φ(m-1)
% m-1
% m

#important thing



$\therefore \varphi(\varphi(\varphi(m)))$

$\therefore \varphi(\varphi(m)$

$\therefore \varphi(m)$

$\therefore m$

Mod Inverse using Euler's Theorem and Fermat's Theorem:

$$a^{\phi(m)} \equiv 1 \mod m \quad \longrightarrow \quad a^{\phi(m)-1} \equiv a^{-1} \mod m$$

$$a^{m-1} \equiv 1 \mod m \quad \longrightarrow \quad a^{m-2} \equiv a^{-1} \mod m$$

Bonus Resources Links [Learn on your own]

- **Extended Euclidean Algo**
  https://cp-algorithms.com/algebra/extended-euclid-algorithm.html
- **Euclidean Algo**
  https://cp-algorithms.com/algebra/euclid-algorithm.html
- **Linear Diophantine Equations**
  https://cp-algorithms.com/algebra/linear-diophantine-equation.html
- **Mod Inverse in O(logM) for general M**
  https://cp-algorithms.com/algebra/module-inverse.html
- **Chinese Remainder Theorem**
  https://cp-algorithms.com/algebra/chinese-remainder-theorem.html
- **Checking for Prime in O(logP) - Miller Rabin Test**
  https://cp-algorithms.com/algebra/primality_tests.html#miller-rabin-primality-test